Research Statement

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1 Background and Current Work

In early fall 2009, at the time when LHC experiments were intensively preparing for the first collisions, I started my PhD thesis work. I began with contributions to the commissioning of the electrons with early data in 2010 and 2011. The reconstruction of electrons in CMS relies on rather elaborate techniques combining information from the electromagnetic calorimeter and the tracker detectors. We first validated the reconstruction of all individual observables and verified the agreement with Monte Carlo (MC) simulation. We then used the very first data with single Z (and W) production and deployed tag-and-probe (T&P) techniques. I co-signed six analysis notes on the topic of electron measurements during that phase.

In the $\mathrm{H} \rightarrow \mathrm{ZZ}^{(\star)} \rightarrow 4$ leptons analysis my contributions include work on definition and implementation of the overall analysis strategy (from the analysis with first data to the most recent results on properties measurements), definition and development of the lepton isolation algorithm, measurements of electrons reconstruction efficiencies, data-to-Monte-Carlo ratios and systematics with the specifically developed T&P method, integration of analysis tools in the software framework, processing and maintaining analysis data samples. In what follows these contributions are explained in more details.

The search for the Higgs boson through its decay to four leptons is known as the "golden channel". It has been considered as one of the flagship channels for analysis in the CMS experiment since the origin of the LHC project. This channel provides the main motivation for high efficiency and precision of lepton reconstruction down to the lowest possible momenta. Such high efficiencies must be achieved while providing powerful lepton identification andisolationobservables for the signal to background discrimination, and allowing for performance and background control from data. The main emphasis of the analysis with first data was the lepton reconstruction and background control. The work was first focused on the deployment of the high efficiency lepton reconstruction algorithm down to low momenta, with the full control of systematic uncertainties. I have participated in the commissioning of electrons with early data, from the overall electron reconstruction to the specific parts, such as charge determination, track seeding, reconstruction efficiency and momentum determination. My work, together with other colleagues from the group, resulted in the fully functional official software for electron reconstruction, which has been used for many analyses in the CMS experiment, and in particular for the Higgs boson search in the four lepton channel.

The framework for the $\mathrm{H} \rightarrow \mathrm{ZZ}^{(\star)} \rightarrow 4$ leptons analysis has been established when my thesis project started. My initial contribution consisted of implementing of the final layer of the analysis work-flow and testing the full analysis chain, including data processing, skimming and maintenance. This part of my contribution has continued until the final published results.

The analysis strategy has evolved with the amount of data collected and has been influenced by our better understanding and control of the detector and the underlying physics. First results of the analysis have been presented in the European Physics Society conference (EPS) in Grenoble in 2011, where we demonstrated excellent control of the lepton reconstruction, understanding of the background processes and robustness of the full analysis chain, including the statistical interpretation of results. The further natural evolution of analysis included opening of the phase space to accept more signal events (developing and deploying more involved tools for background control in constantly increasing hostile environment of more and more pile-up events) and to explore full event kinematics and sophisticated analysis methods. My contribution to this process was in developing, deploying and testing the analysis tools, integrating the full analysis chainand optimizing the selection steps, documenting and presenting the results in internal and external meetings.

A particular emphasis throughout all my PhD has been put on lepton isolation as a key ingredient in our Higgs boson search. The lepton isolation is one of the key observables for the Higgs search in 4 lepton channel. Nevertheless, it has to be carefully designed to take into account the kinematics of the Higgs decay (spin 0) to Zs (spin 1) and further on to leptons (spin 1/2) . In many cases (~5%) leptons from either Z are quasi-collinear, thus entering into each others isolation cone. This happens for Higgs boson at low mass since Zs are produced at rest but also for Higgs boson at significantly high masses when Zs acquire boost. Part of my work was dedicated to properly exclude the nearby lepton energy deposit when computing isolation to avoid loosing the efficiency of selection. Continuing to work with lepton isolation, I participated in pile-up study task force in 2011 to check the impact of the multiple interactions on isolation observables. Isolation is, of course, susceptible to pile-up, when calorimetry is considered because the information on vertex is not available. I worked on establishing an isolation calculation method immune to additional energy flow from pile-up interactions. The method is known across CMS as the "effective area" correction (EA). It uses the information of the average energy density in detector obtained via FastJet calculation to estimate the pile-up. Using calorimeter instead of simple vertex multiplicity information gives a handle also to out-of-time pile-up. Since one of the 4 leptons in the final state typically has a transverse momentum of less than 10 GeV, we had to push the lepton acceptances to values as low as 5 (7) GeV for muons (electrons). In this low p_T region, the data-to-MC discrepancies are expected to be larger, therefore a solid data driven control of efficiencies had to be established. I was directly working on this issue using tag-and-probe technique which profits from leptonic Z decays to evaluate selection efficiencies directly from data. These measurements were used in the analysis in the form of per-lepton scale factors and its uncertainties propagate trough the analysis.

Few months before my thesis defence, in parallel with writing the thesis, I started the work on CMS Level-1 calorimetric trigger Phase 2 upgrade in view of preparation of the CMS Level-1 Upgrade Technical Design Report. The goal of these studies were the determination of the expected performance of electron (photon) and tau triggering algorithms, as well as their optimization and development. Using collision data from Run 1, the performances were also directly compared to the current trigger system. Given my previous experience on the multi-lepton channels ($\mathrm{H} \rightarrow \mathrm{ZZ}^{(\star)} \rightarrow 4$ leptons) as well as lepton reconstruction, identification and selection, I managed to immerse quickly into this more detector oriented subject. This work has been carried out in collaboration with Alexandre Zabi from LLR, École Polytechnique (France) and published within the Level-1 Trigger Upgrade TDR.

Currently, I am a postdoc at IHEP, Beijing, China. My main project is the development of cross-check analysis for the Higgs boson properties measurements. This study is still ongoing. In addition take part in the measurement of Higgs boson fiducial cross-section which is currently in preapproval phase in CMS collaboration. This effort is still ongoing in close collaboration of University of Florida, University of Virginia and IHEP - Beijing teams.

2 Future Research Interests

Having in mind the proximity of the LHC Run 2, I think it crucial to prepare the $H \rightarrow 4l$ analysis to:

- Cope with high pile-up,
- Use of new trigger paths,
- Include new object developments,
- Optimize the selection strategy through categorization (production modes, number of leptons),
- Use more advanced techniques (e.g. MVA).

In addition to preparation of Run 2, my high priority interest is to continue the Higgs boson anomalous couplings studies using the MEKD (Matrix Element Kinematic Discriminant) framework developed by UF group at CMS. New data from Run 2 will allow for more precise couplings measurements and will give as more information on possible new physics via small deviations from Standard Model Higgs boson couplings structure.

In contrary with the high-level physics analysis interests, my short but sweet experience with $e/\gamma/\tau$ Level-1 trigger provides me with enthusiasm to carry out the work related to more basic detector aspects. In order to broaden my research experience, I am keen to learn, work and provide service in the CSC Level-1 trigger area of expertise.